

gate_crushers

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Quantum Hack 26

One reproducible QEC pipeline

Same experiment definition runs in calibrated simulation or on IQM hardware.



Artifacts for every run: circuit.stim, Qiskit circuits, counts, raw measurements, detector events, syndrome diagnostics, metrics, and plots.

Why this matters: hardware data can be decoded again offline, enabling fair decoder comparisons without spending more QPU time.

Hardware calibration enters two places

Spatial calibration is used for patch selection; per-qubit/per-coupler errors drive calibrated simulation.

PLACEMENT PATH

IQM observation dump
PRX · readout · QND · T2 ·
CZ

**Best native patch
or routed layout**

**Qiskit initial_layout
→ transpile**

NOISE MODEL PATH

**Mapped qubits
and couplers**

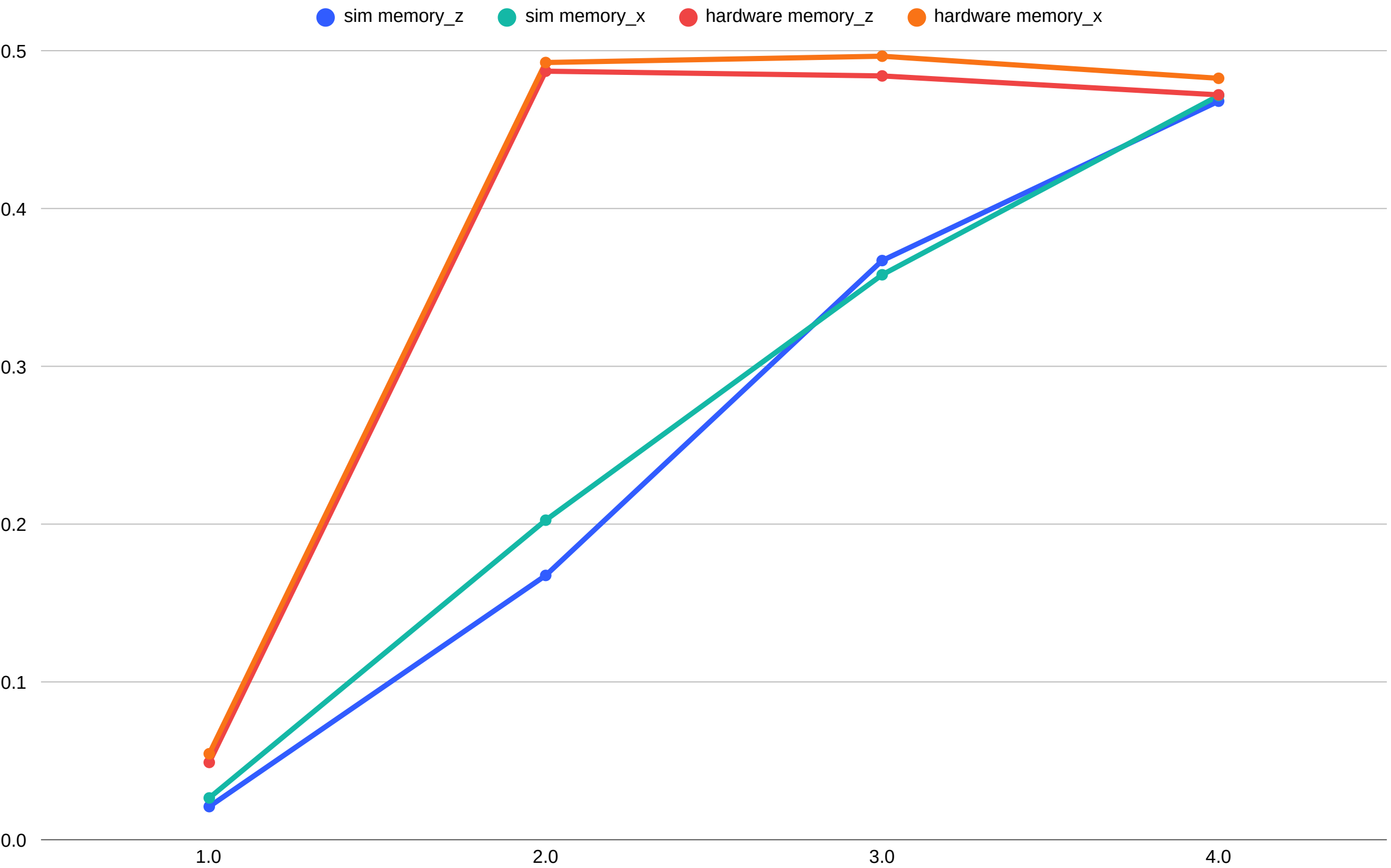
Stim noise injection
PRX · CZ · readout · idle

**Calibrated simulator
LER vs rounds**

Key caveat: Stim-level calibration is useful but does not model pulse timing, leakage, crosstalk, drift, or repeated reset/readout history.

Simulator predicts gradual decay; hardware saturates early

Distance-3, 2000 shots, paired simulator/hardware sweep on the calibrated Emerald patch.



≈0.05

hardware round 1

≈0.49

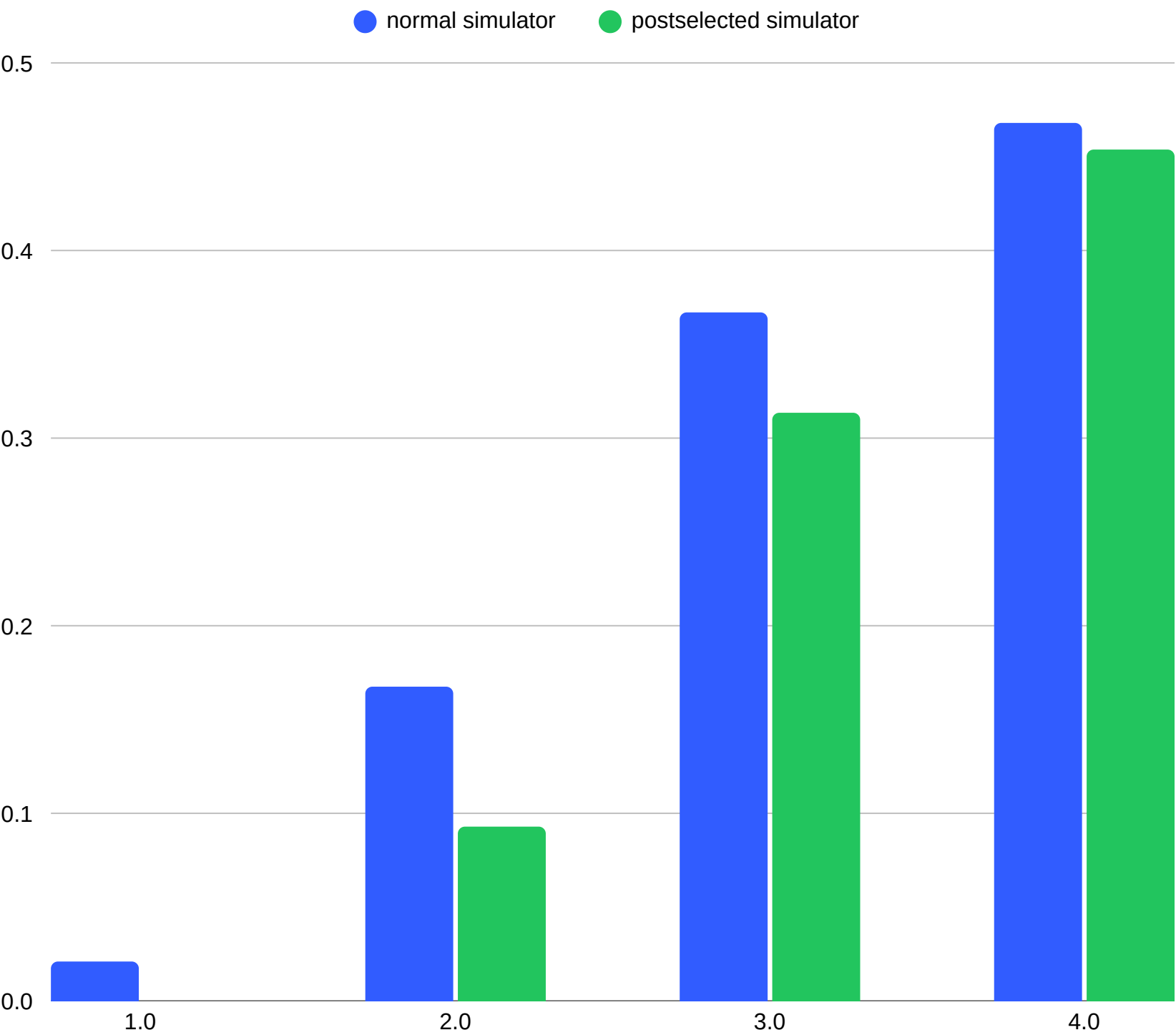
hardware round 3+

Interpretation: the one-round hardware result is useful, but repeated-round behavior is not captured by the current Stim-level calibration model.

LER near 0.5 means the decoded logical outcome is effectively random for one logical observable.

Improvement experiments: what moved the needle

We tested circuit variants, patch selection, decoder routes, and low-syndrome postselection.



WHAT WE TRIED

- **d=3 / d=5** rotated and unrotated surface-code routes
- **Z and X memory** paired basis experiments
- **Calibration mapping** native d=3 patch; routed d=5 layout
- **Reset variants** initial-reset and repeated-reset A/B tests
- **Decoder routes** MWPM, auto-selection, postselection

Postselection is reported honestly: it lowers LER on accepted shots, not the underlying physical failure rate.

Main finding: model mismatch appears only in repeated rounds

This is the most useful scientific signal from the project.

Round 1 works

hardware $\approx$ simulator order of magnitude

Rounds 3+ saturate

hardware LER ≈ 0.49

Missing physics

beyond Stim-level calibration

LIKELY MISSING EFFECTS

repeated reset/readout dynamics

measurement back-action

leakage

crosstalk

timing / scheduling overhead

pulse-level compilation effects

What we would do next

The pipeline is ready for targeted hardware-level and decoder-level follow-up.

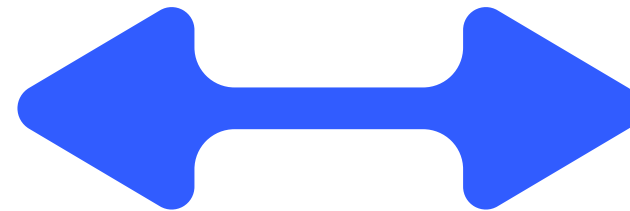
- 1 Run paired hardware postselection**
Check if low-syndrome filtering improves real round-3 LER.
- 2 Pulse-level / DD path**
Reduce idle and scheduling errors before syndrome extraction.
- 3 Repeated-round noise fitting**
Fit reset/readout history instead of one-round calibration only.
- 4 Learned or calibration-weighted decoder**
Use saved hardware shots to test decoder improvements offline.

Takeaway: we built a reproducible QEC experiment framework that reveals where calibrated simulation stops matching real repeated-round hardware.

The Discovery That Changed Our Direction

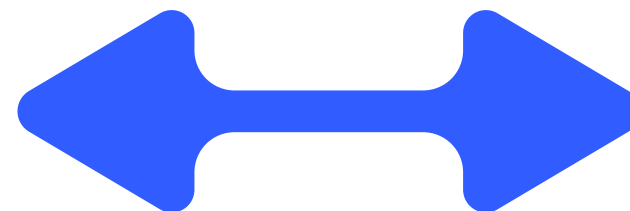
The most valuable result was not a lower error rate—it was finding where the model breaks.

Round 1: Hardware and calibrated simulation agree.



Rounds 3+: Hardware logical error rate saturates near 0.49.

Simulation predicts gradual decay instead.



The mismatch appears only during repeated syndrome cycles.

Why This Matters for IQM

From benchmarking performance to diagnosing hardware limitations.

**Reproducible QEC
pipeline across
simulation and
hardware.**

**Offline decoder
comparisons
without additional
QPU time.**

**Direct
identification of
missing physical
effects.**

**A practical
workflow for future
hardware
development.**

Final Takeaway

A framework for discovering where calibrated models stop matching reality.

